RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year - Semester II Examination – March - May 2026

COM3306 – OPERATING SYSTEMS

Time: Three (03) hours

- There are **Five (05)** questions in **Three (03)** pages.
- Answer **ALL** questions.

1. a) State three (03) main components of a computer system. (03 marks)

   b) Explain the significance of the separation between user mode and kernel mode in operating systems. (04 marks)

   c) Illustrate how an operating system running on a uni-processor system manages the execution of multiple application programs in a multitasking environment. (06 marks)

   d) Discuss how operating systems contribute to manage the performance of the system and user convenience (07 marks)

2. a) List the basic states in the life cycle of a process in the context of operating systems. (03 marks)

   b) Compare the roles of the short-term scheduler and the long-term scheduler in process management. (04 marks)

   c) Analyze how process queues contribute to CPU and I/O resource allocation in an operating system. (05 marks)

d) The following table indicates the arrival time and CPU burst time of five processes P1, P2, P3, P4, and P5.

Process	Arrival Time(ms)	Burst Time (ms)
P1	0	7
P2	2	4
P3	4	1
P4	5	4
P5	6	6

i. Construct the Gantt charts for the FCFS and Round Robin (Time Quantum = 3 ms) scheduling algorithms. (04 marks)

ii. Find the average completion time, average waiting time, and average turnaround time for each of the above scheduling algorithms. (02 marks)

iii. Evaluate which of the above two scheduling algorithms is more suitable for an interactive computing environment. (02 marks)

3. a) Explain the concept of race conditions in process synchronization. (03 marks)

b) Compare the lock variable mechanism and Peterson's solution as approaches to process synchronization. (04 marks)

c) Analyze how the critical section problem affects the correctness and consistency of concurrent process execution. (05 marks)

d) Evaluate how synchronization mechanisms contribute to both process coordination and system security in a multi-user operating system. (08 marks)

4. a) Briefly explain the concept of deadlock in operating systems and describe its impact on process execution. (03 marks)

b) Differentiate between deadlock prevention and deadlock avoidance as strategies for handling deadlock. (04 marks)

c) Discuss the significance of the four (04) necessary conditions required for deadlock occurrence. (05 marks)

d) The following tables show the resource allocation state of a system with five processes (P0 - P4) and three resource types A, B, and C. Apply Banker's Algorithm to determine whether the system is in a safe state. If the system is safe, provide a safe sequence.
Available vector: (A-3, B-3, C-2) (08 marks)

Allocation matrix

Process	A	B	C
P0	0	1	0
P1	2	0	0
P2	3	0	2
P3	2	1	1
P4	0	0	2

Maximum Matrix

Process	A	B	C
P0	7	5	3
P1	3	2	2
P2	9	0	2
P3	2	2	2
P4	4	3	3

5. a) State three (03) different levels of memory hierarchy and give one (01) key characteristic of each. (03 marks)

b) Describe static and dynamic loading in memory management and explain how dynamic loading improves memory usage during program execution. (04 marks)

c) Analyze how paging and segmentation solve the problems of contiguous memory allocation. Discuss one (01) advantage and one (01) disadvantage of each method. (06 marks)

d) In a multi-user operating system, important files are stored for different users. Evaluate how the direct access method, tree, and Directed Acyclic Graph (DAG) directory structures help prevent data conflicts, improve efficiency, and maintain security. (07 marks)

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